Sound

Sound Production

Sound production consists of four phases: design, recording, editing, and mixing. Although we might suppose that most of the sounds in a movie are the result of recording during filming (such sounds are called *production sounds*), the reality is that most film sounds are constructed during the postproduction phase (and thus are called *postproduction sounds*). But before any sounds are recorded or constructed, the overall plan for a movie's sound must be made. That planning process is called *sound design*.

Design

Sound design is the art of creating the sound for a film. As motion-picture sound has become increasingly innovative and complex, the result of comprehensive sound design, the sound designer's role has become more well known. Given its name by film editor Walter Murch, sound design combines the crafts of editing and mixing and, like them, involves matters both theoretical and practical.

Although many filmmakers continue to understand and manipulate sound in conventional ways, sound design has produced major advances in how movies are conceived, made, viewed, and interpreted. Until the 1970s, the vast majority of producers and directors thought about sound only after the picture was shot. They did not design films with sound in mind and frequently did not fully recognize that decisions about art direction, composition, lighting, cinematography, and acting would ultimately influence how sound tracks would be created and mixed. They considered sound satisfactory if it could distract from or cover up mistakes in shooting and create the illusion that the audience was hearing what it was seeing.

By contrast, the contemporary concept of sound design rests on the following basic assumptions:

- Sound should be integral to all three phases of film production (preproduction, production, and postproduction), not an afterthought to be added in postproduction only.
- A film's sound is potentially as expressive as its images.
- Image and sound can create different worlds.
- Image and sound are co-expressible.

A sound designer treats the sound track of a film the way a painter treats a canvas. For each shot, the designer first identifies all the sounds necessary to the story and plot. The next step is laying in all the background tones (different tones equal different colors) to create the support necessary for adding the specific sounds that help the scene to function.

Before sound design was widely accepted, the responsibilities for sound were divided among recording, rerecording, editing, mixing, and sound-effects crews; these crews sometimes overlapped but often did not. In the industry's attempt to integrate all aspects of sound in a movie, from planning to postproduction, the sound designer began to supervise all these responsibilities—a development initially resented by many traditional sound specialists, who felt their autonomy was being compromised. It is now conventional for sound designers (or supervising sound editors) to oversee the creation and control of the sounds (and silences) we hear in movies. They are, in a sense, advocates for sound.

Recording

The process of recording sound is very similar to the process of hearing. Just as the human ear converts sounds into nerve impulses that the brain identifies, so the microphone converts sound waves into electrical signals that are then recorded and stored. The history of recording movie sound has evolved from optical and magnetic systems to the digital systems used in today's professional productions. Of the various types of film sound (which will be described later in the chapter), dialogue is the only type typically recorded during production. Everything else is added in the editing and mixing stages of postproduction.

The recording of production sound is the responsibility of the production sound mixer and a team of assistants, which includes, on the set, a sound recordist, a sound mixer, a microphone boom operator, and wranglers (in charge of the power supply, electrical connections, and cables).

On set, the motion-picture camera is responsible only for recording the image; the dialogue sound is recorded using a separate sound recorder, an approach known as double-system recording. Before any dialogue shot is captured on set, an assistant "claps" the hinged pieces of a simple device called the slate (also known as a clapboard or sticks) to create a simultaneous image and sound "mark" that are used to line up (or synchronize) the separate image and sound recordings in postproduction, a process referred to as synching.

Editing

The editor is responsible for the overall process of editing and for the sound crew, which consists of a supervising sound editor, sound editors (who usually concentrate on their specialties: dialogue, music, or sound effects), sound mixers, rerecording mixers, sound-effects personnel, and Foley artists. The editor also works closely with the musical composer or those responsible for selecting music from other sources. In the editing room, the editor is in charge; but the director and the sound designer may also take part in the process.

Filmmakers first screen the dailies (or rushes), which are synchronized picture/sound work prints of a day's shooting. From these they select the usable individual shots from among the multiple takes, sort out the outtakes (any footage that will not be used), log the usable footage so it is easy to follow through the rest of the process, and decide which dialogue needs rerecording and which sound effects are necessary. If ambient or other noises have marred the quality of the dialogue recorded during photography, the actors are asked to come back, view the faulty scene, and perform the dialogue again while watching a looped (repeating) recording of the moment in question, a process known as automatic dialogue replacement (ADR), or looping. It's very much like selective lip-synching; when an acceptable rerecording that matches the take has been made, an ADR editor inserts it into the movie. Finally, the sound-editing team synchronizes the sound and visual tracks.

Automatic Dialogue Replacement (ADR)

In this post-production session for *Blood Stripe* (2016, director Remy Auberjonois), actor Kate Nowlin uses ADR to replace dialogue she delivered on set. Watching a repeating (or

looped) clip of her original performance allows the actor to record multiple takes until she achieves the synchronization needed for a convincing new vocal performance. Automatic (or automated) dialogue replacement is used to adjust dialogue delivery or replace flawed recordings.

Mixing

Mixing is the process of combining all of the different individual edited tracks of dialogue, sound effects, music, and so forth, into one composite sound track to play in synchronization with the edited picture.

The number of sound tracks used in a movie depends on the kind and amount of sound needed to tell each part of the story; thus, filmmakers have an unlimited resource at their disposal. No matter how many tracks are used, they are usually combined and compressed during the final mixing. Working with their crew, sound mixers adjust the relative loudness and various aspects of sound quality; filter out unwanted sounds; and create, according to the needs of the screenplay, the right balance of dialogue, music, and sound effects. The result is a sort of “audio mise-en-scène” that emphasizes significant sound elements in the mix, just as a visual composition uses placement and size in frame to feature significant subject matter in a shot.